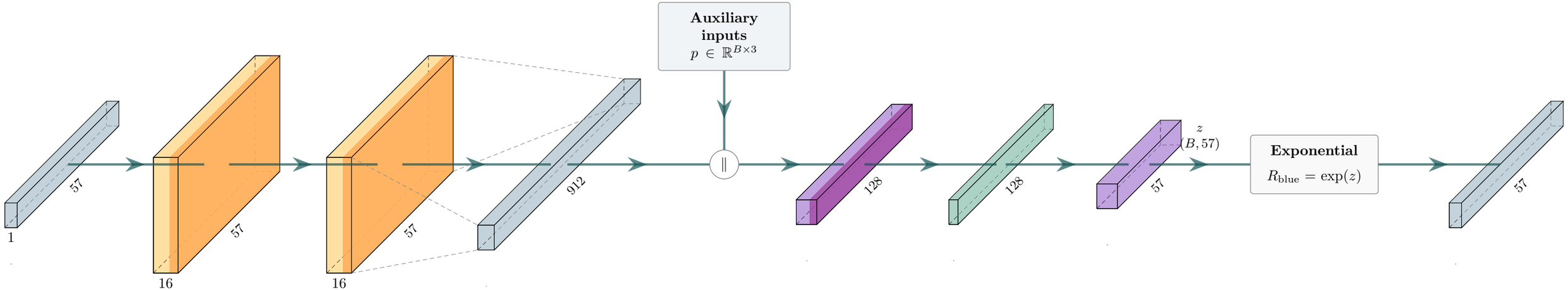
# BlueWiggleNet

$N_b = 57$ blue wavelength bins  $P = 3$ auxiliary inputs  $H = 128$

`forward(x_blue, p)`: **pixel-wise multiplicative correction**




**Normalized input**
$x_{\text{norm}}$

**Conv1d + GELU**
$k = 3,\ s = 1,\ p = 1$

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**Flatten**
$16 \times 57 = 912$

**Concatenate**
$912 + 3 = 915$

**Linear + GELU**
$915 \to 128$

**Dropout**
$p = 0.2$
training only

**Linear**
$128 \to 57$

**Element-wise**
positive ratios

**Blue correction**
$R_{\text{blue}} \in \mathbb{R}^{B\times 57}_{>0}$

**Input normalization** (original Gaia blue flux)

$s = \max_\lambda |x_{\text{blue}}| + 10^{-20}, \quad x_{\text{norm}} = \text{unsqueeze}(x_{\text{blue}}/s)$

**Tensor labels** (batch axis $B$ omitted)

Feature maps: channels and blue wavelength bins; vectors: features.
Convolutions preserve wavelength sampling; block sizes are schematic.

**Output semantics**

Final linear layer: $W = 0$, $b = 0$ initially.
Hence $z = 0$ and $R_{\text{blue}} = 1$: the initial correction is neutral.

**Legend** Conv1d  Linear  Dropout  Bands: GELU

Auxiliary inputs: BP−RP, G, RUWE; standardized before the model.